

Abhijeet Anand

AbhijeetAnand@lbl.gov, <https://abh0395.github.io>

Postdoctoral Scientist, Lawrence Berkeley National Lab, Berkeley, CA

Positions

Sep'22 - Aug'25: Postdoctoral Scientist, Lawrence Berkeley National Lab (LBNL), Berkeley, CA, USA

Aug'22: Bridge Postdoctoral Fellow, Max Planck Institute for Astrophysics, Garching, Germany

Sep'18 - Jul'22: IMPRS PhD Fellow, Max Planck Institute for Astrophysics, Garching, Germany

Education

Sep 2018 - Jul 2022: PhD in Astrophysics, Ludwig Maximilian University - Max Planck Institute for Astrophysics (MPA), Germany.

- Thesis: [Probing cool and warm circumgalactic gas in galaxies and clusters with large spectroscopic and imaging surveys](#)
- Advisors: [Prof. Dr. Guinevere Kauffmann](#) & [Dr. Dylan Nelson](#)

Jul 2016 - Jul 2017: MS, Physics, Indian Institute of Science (IISc), Bangalore, India.

- Thesis: [A sensitive search for HI 21 cm emission from super disks in radio galaxies](#)
- Advisor: [Prof. Nirupam Roy](#)

Aug 2012 - June 2016: BSc (Research), Physics, Indian Institute of Science, Bangalore, India.

- Thesis: [Sources of Continuum Opacity in Hydrogen deficient stars](#)
- Advisors: [Prof. Gajendra Pandey](#), Indian Institute of Astrophysics (IIA)

Publications

- ³² Moore, Samuel G.; Cole, Shaun; Wilson, Michael; Norberg, Peder; Moustakas, John; *et al.* (incl. **A. Anand**), 2026, *DESI DR2 Galaxy luminosity functions*, Monthly Notices of the Royal Astronomical Society, **549**, stag866 ([arXiv:2511.01803](#)) [6 citations]
- ³¹ DESI Collaboration; Abdul Karim, M.; Adame, A. G.; Aguado, D.; Aguilar, J.; *et al.* (incl. **A. Anand**), 2026, *Data Release 1 of the Dark Energy Spectroscopic Instrument*, The Astronomical Journal, **171**, 285 ([arXiv:2503.14745](#)) [447 citations]
- ³⁰ Turner, Wynne; Cuceu, Andrei; Martini, Paul; Aguilar, J.; Ahlen, S.; *et al.* (incl. **A. Anand**), 2026, *Probing the limits of cosmological information from the Lyman- α forest 2-point correlation functions*, Journal of Cosmology and Astroparticle Physics, **2026**, 010 ([arXiv:2509.14322](#)) [3 citations]
- ²⁹ Zaborowski, E. A.; Taylor, P.; Honscheid, K.; Cuceu, A.; de Mattia, A.; *et al.* (incl. **A. Anand**), 2026, *$H <_{SUB} 0 </SUB>$ without the sound horizon (or supernovae): A 2% measurement in DESI DR1*, Journal of Cosmology and Astroparticle Physics, **2026**, 004 ([arXiv:2510.19149](#)) [4 citations]
- ²⁸ Karaçaylı, Naim Göksel; Ravoux, Corentin; Martini, Paul; Goff, Jean-Marc Le; Armengaud, Eric; *et al.* (incl. **A. Anand**), 2026, *DESI DR1 Ly α 1D power spectrum: Validation of estimators*, Journal of Cosmology and Astroparticle Physics, **2026**, 048 ([arXiv:2509.13593](#)) [4 citations]
- ²⁷ Ebina, H.; White, M.; Raichoor, A.; Dey, Arjun; Schlegel, D.; *et al.* (incl. **A. Anand**), 2026, *Clustering analysis of medium-band selected high-redshift galaxies*, Journal of Cosmology and Astroparticle Physics, **2026**, 019 ([arXiv:2509.26467](#)) [3 citations]

- 26 Hsu, Yuan-Ming; Huang, Xiaosheng; Storfer, Christopher J.; Inchausti, Jose Carlos; Schlegel, David; *et al.* (incl. **A. Anand**), 2026, *A New Way to Discover Strong Gravitational Lenses: Pairwise Spectroscopic Search from DESI DR1*, The Astrophysical Journal Supplement Series, **282**, 41 (arXiv:2509.16033) [2 citations]
- 25 Herrera-Alcantar, Hiram K.; Armengaud, Eric; Yèche, Christophe; Gordon, Calum; Casas, Laura; *et al.* (incl. **A. Anand**), 2025, *The Lyman- α forest from LBGs: First 3D correlation measurement with DESI and prospects for cosmology*, Journal of Cosmology and Astroparticle Physics, **2025**, 053 (arXiv:2507.21852) [4 citations]
- 24 Abdul Karim, M.; Aguilar, J.; Ahlen, S.; Alam, S.; Allen, L.; *et al.* (incl. **A. Anand**), 2025, *DESI DR2 results. II. Measurements of baryon acoustic oscillations and cosmological constraints*, Physical Review D, **112**, 083515 (arXiv:2503.14738) [1164 citations]
- 23 Abdul Karim, M.; Aguilar, J.; Ahlen, S.; Allende Prieto, C.; Alves, O.; *et al.* (incl. **A. Anand**), 2025, *DESI DR2 results. I. Baryon acoustic oscillations from the Lyman alpha forest*, Physical Review D, **112**, 083514 (arXiv:2503.14739) [223 citations]
- 22 Brodzeller, A.; Wolfson, M.; Santos, D. M.; Ho, M.; Tan, T.; *et al.* (incl. **A. Anand**), 2025, *Construction of the damped Ly α absorber catalog for DESI DR2 Ly α BAO*, Physical Review D, **112**, 083510 (arXiv:2503.14740) [21 citations]
- 21 Rashkovetskyi, Michael; Eisenstein, Daniel J.; Aguilar, Jessica Nicole; Ahlen, Steven; **Anand, Abhijeet**; *et al.*, 2025, *Clustering of DESI galaxies split by thermal Sunyaev-Zeldovich effect*, The Open Journal of Astrophysics, **8**, 46033 (arXiv:2508.20904)
- 20 Adame, A. G.; Aguilar, J.; Ahlen, S.; Alam, S.; Alexander, D. M.; *et al.* (incl. **A. Anand**), 2025, *DESI 2024 V: Full-Shape galaxy clustering from galaxies and quasars*, Journal of Cosmology and Astroparticle Physics, **2025**, 008 (arXiv:2411.12021) [134 citations]
- 19 **Anand, Abhijeet**; Aguilar, J.; Ahlen, S.; Bianchi, D.; Brodzeller, A.; *et al.*, 2025, *The Cosmic Evolution of C IV Absorbers at $1.4 < z < 4.5$: Insights from 100,000 Systems in DESI Quasars*, The Astrophysical Journal, **990**, 151 (arXiv:2504.20299) [6 citations]
- 18 Adame, A. G.; Aguilar, J.; Ahlen, S.; Alam, S.; Alexander, D. M.; *et al.* (incl. **A. Anand**), 2025, *DESI 2024 VII: cosmological constraints from the full-shape modeling of clustering measurements*, Journal of Cosmology and Astroparticle Physics, **2025**, 028 (arXiv:2411.12022) [312 citations]
- 17 Adame, A. G.; Aguilar, J.; Ahlen, S.; Alam, S.; Alexander, D. M.; *et al.* (incl. **A. Anand**), 2025, *DESI 2024 II: sample definitions, characteristics, and two-point clustering statistics*, Journal of Cosmology and Astroparticle Physics, **2025**, 017 (arXiv:2411.12020) [119 citations]
- 16 Adame, A. G.; Aguilar, J.; Ahlen, S.; Alam, S.; Alexander, D. M.; *et al.* (incl. **A. Anand**), 2025, *DESI 2024 III: baryon acoustic oscillations from galaxies and quasars*, Journal of Cosmology and Astroparticle Physics, **2025**, 012 (arXiv:2404.03000) [503 citations]
- 15 Wu, Xuanyi; Cai, Z.; Lan, T.-W.; Zou, S.; **Anand, Abhijeet**; *et al.*, 2025, *Tracing the Evolution of the Cool Gas in CGM and IGM Environments through Mg II Absorption from Redshift $z = 0.75$ to $z = 1.65$ Using DESI-Y1 Data*, The Astrophysical Journal, **983**, 186 (arXiv:2407.17809) [12 citations]
- 14 Adame, A. G.; Aguilar, J.; Ahlen, S.; Alam, S.; Alexander, D. M.; *et al.* (incl. **A. Anand**), 2025, *DESI 2024 VI: cosmological constraints from the measurements of baryon acoustic oscillations*, Journal of Cosmology and Astroparticle Physics, **2025**, 021 (arXiv:2404.03002) [1630 citations]
- 13 Adame, A. G.; Aguilar, J.; Ahlen, S.; Alam, S.; Alexander, D. M.; *et al.* (incl. **A. Anand**), 2025, *DESI 2024 IV: Baryon Acoustic Oscillations from the Lyman alpha forest*, Journal of Cosmology

- and Astroparticle Physics, **2025**, 124 (arXiv:2404.03001) [365 citations]
- 12 Ross, A. J.; Aguilar, J.; Ahlen, S.; Alam, S.; **Anand, Abhijeet**; et al., 2025, *The construction of large-scale structure catalogs for the Dark Energy Spectroscopic Instrument*, Journal of Cosmology and Astroparticle Physics, **2025**, 125 (arXiv:2405.16593) [53 citations]
 - 11 Scholte, Dirk; Saintonge, Amélie; Moustakas, John; Catinella, Barbara; Zou, Hu; et al. (incl. **A. Anand**), 2024, *The atomic gas sequence and mass-metallicity relation from dwarfs to massive galaxies*, Monthly Notices of the Royal Astronomical Society, **535**, 2341 (arXiv:2408.03996) [26 citations]
 - 10 Chang, Yu-Ling; Lan, Ting-Wen; Prochaska, J. Xavier; Napolitano, Lucas; **Anand, Abhijeet**; et al., 2024, *Probing the Impact of Radio-mode Feedback on the Properties of the Cool Circumgalactic Medium*, The Astrophysical Journal, **974**, 191 (arXiv:2405.08314) [8 citations]
 - 9 Galiullin, Ilkham; Rodriguez, Antonio C.; El-Badry, Kareem; Szkody, Paula; **Anand, Abhijeet**; et al., 2024, *Searching for new cataclysmic variables in the Chandra Source Catalog*, Astronomy and Astrophysics, **690**, A374 (arXiv:2408.00078) [6 citations]
 - 8 **Anand, Abhijeet**; Guy, Julien; Bailey, Stephen; Moustakas, John; Aguilar, J.; et al., 2024, *Archetype-based Redshift Estimation for the Dark Energy Spectroscopic Instrument Survey*, The Astronomical Journal, **168**, 124 (arXiv:2405.19288) [41 citations]
 - 7 DESI Collaboration; Adame, A. G.; Aguilar, J.; Ahlen, S.; Alam, S.; et al. (incl. **A. Anand**), 2024, *The Early Data Release of the Dark Energy Spectroscopic Instrument*, The Astronomical Journal, **168**, 58 (arXiv:2306.06308) [534 citations]
 - 6 DESI Collaboration; Adame, A. G.; Aguilar, J.; Ahlen, S.; Alam, S.; et al. (incl. **A. Anand**), 2024, *Validation of the Scientific Program for the Dark Energy Spectroscopic Instrument*, The Astronomical Journal, **167**, 62 (arXiv:2306.06307) [295 citations]
 - 5 Napolitano, Lucas; Pandey, Agnesh; Myers, Adam D.; Lan, Ting-Wen; **Anand, Abhijeet**; et al., 2023, *Detecting and Characterizing Mg II Absorption in DESI Survey Validation Quasar Spectra*, The Astronomical Journal, **166**, 99 (arXiv:2305.20016) [28 citations]
 - 4 Ayromlou, Mohammadreza; Kauffmann, Guinevere; **Anand, Abhijeet**; & White, Simon D. M., 2023, *The physical origin of galactic conformity: from theory to observation*, Monthly Notices of the Royal Astronomical Society, **519**, 1913 (arXiv:2207.02218) [29 citations]
 - 3 **Anand, Abhijeet**; Kauffmann, Guinevere; & Nelson, Dylan, 2022, *Cool circumgalactic gas in galaxy clusters: connecting the DESI legacy imaging survey and SDSS DR16 Mg II absorbers*, Monthly Notices of the Royal Astronomical Society, **513**, 3210 (arXiv:2201.07811) [29 citations]
 - 2 **Anand, Abhijeet**; Nelson, Dylan; & Kauffmann, Guinevere, 2021, *Characterizing the abundance, properties, and kinematics of the cool circumgalactic medium of galaxies in absorption with SDSS DR16*, Monthly Notices of the Royal Astronomical Society, **504**, 65 (arXiv:2103.15842) [61 citations]
 - 1 **Anand, Abhijeet**; Roy, Nirupam; & Gopal-Krishna, 2019, *Search for H I emission from superdisk candidates associated with radio galaxies*, Research in Astronomy and Astrophysics, **19**, 083 (arXiv:1812.06875) [3 citations]